

Approximating the Average-Case Graph Search Problem with Non-Uniform Costs

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Binary Search

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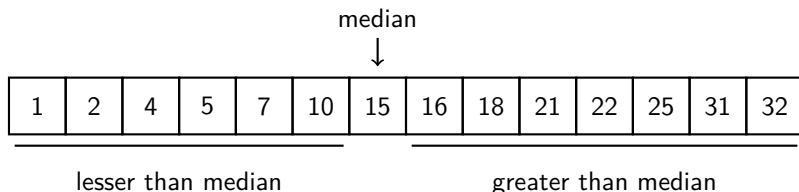


Figure: Example of a sorted array containing 14 elements.

Searching in Graphs

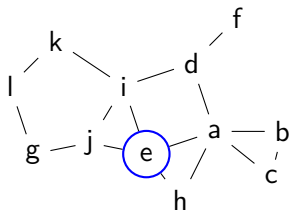
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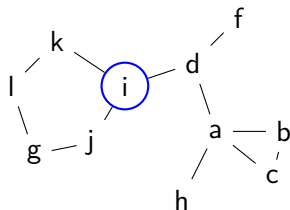
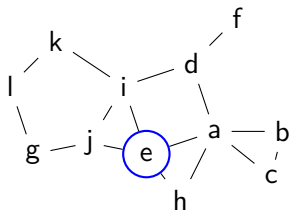
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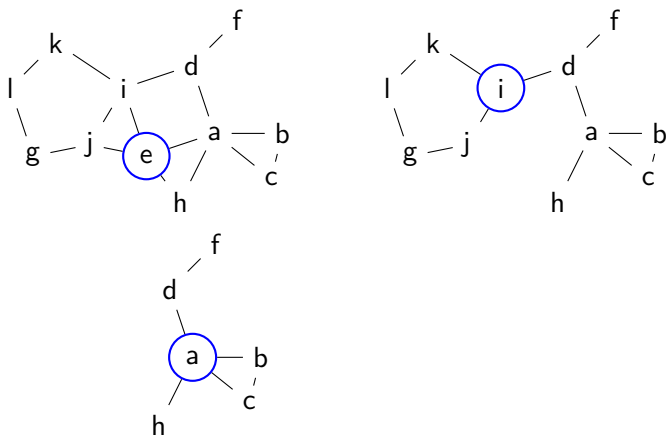
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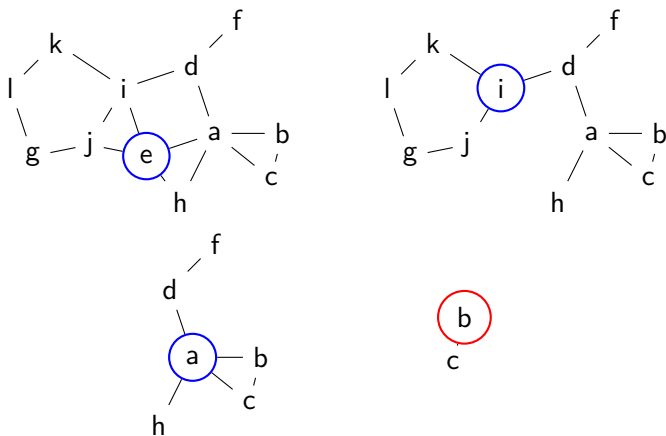
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Strategy of searching

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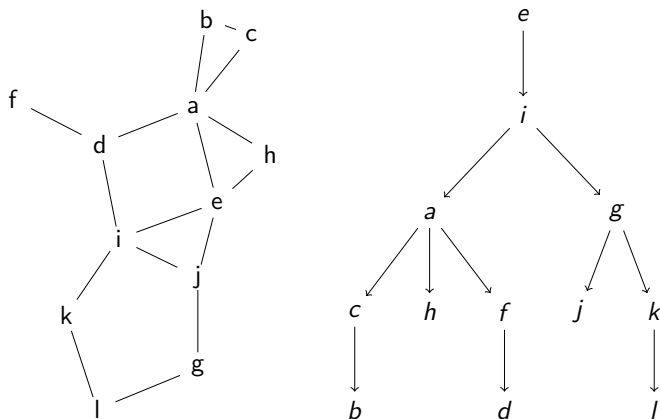


Figure: Sample input graph a decision tree for it.

Additional parameters

We introduce additional information about the input:

- ▶ To each vertex v we assign an arbitrary **cost** $c(v)$, which denotes a cost of performing a query to v .

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- ▶ To each vertex v we assign an arbitrary **cost** $c(v)$, which denotes a cost of performing a query to v .
- ▶ Additionally, to each vertex v we also assign an arbitrary **weight** $w(v)$, which denotes the importance of v .
- ▶ For any $S \subseteq V(G)$, we denote $c(S) = \sum_{v \in S} c(v)$ and $w(S) = \sum_{v \in S} w(v)$.

Our setup

What is the best strategy of searching in a graph?

Graph Search Problem (GSP)

Input: Graph G , a query cost function $c: V(G) \rightarrow \mathbb{N}$ and a weight function $w: V(G) \rightarrow \mathbb{N}$.

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Input: Graph G , a query cost function $c: V(G) \rightarrow \mathbb{N}$ and a weight function $w: V(G) \rightarrow \mathbb{N}$.

Output: A decision tree D minimizing the weighted average search cost:

$$c_G(D) = \sum_{x \in V(G)} w(x) \cdot c(Q_G(D, x))$$

where $Q_G(D, x)$ denotes the set of queries performed along the unique path in D from the root $r(D)$ to x .

Decision Tree

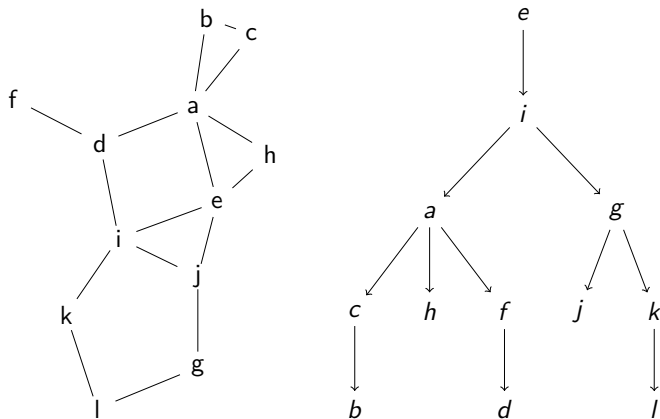


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Other variants

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- ▶ $T || V || \sum w_i C_i$:
 - ▶ $(1 + \epsilon)$ in $O\left((1/\epsilon)^{2/\log_2 3} \cdot n^{1+4/\log_2 3} \cdot \log^2(n/\epsilon)\right)$ time.
 - ▶ Every optimal decision tree has height at most $O(\log w(T))$.
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Bad news: The Graph Search Problem is **NP-hard** even when restricted to bounded degree trees and bounded diameter trees. We want to find an algorithm providing a good **approximation** for the problem:

- ▶ $(4 + \epsilon)$ -approximation for **trees**.
- ▶ $O(\sqrt{\log n})$ -approximation for **general graphs**.

Weighted α -Separator Problem

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Input: Graph G , a cost function $c: V \rightarrow \mathbb{N}$, a weight function $w: V \rightarrow \mathbb{N}$ and a real number α .

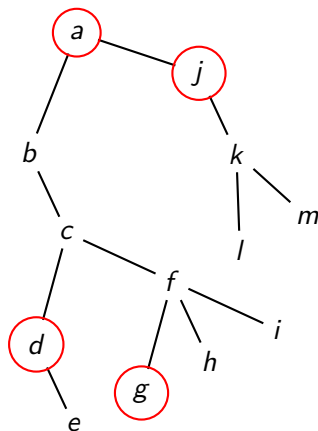
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Input: Graph G , a cost function $c: V \rightarrow \mathbb{N}$, a weight function $w: V \rightarrow \mathbb{N}$ and a real number α .

Output: A set $S \subseteq V(G)$ called **separator** such that for every $H \in G - S$, $w(H) \leq w(G)/\alpha$ and $c(S)$ is minimized.

Example of a separator



	$w(v)$	$c(v)$
<i>a</i>	2	3
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<i>c</i>	3	6
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<i>l</i>	2	3
<i>m</i>	3	4

Figure: Sample input tree T and a weighted 3-separator (the circled vertices) of cost 8.

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3. The algorithm runs in $O(n^3/\delta^2)$ time.

Notation

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- ▶ $S_k^* = V(G) - \mathcal{L}_k^*$ – vertices belonging to the separator at the level $\mathcal{L}_k^*$. S_k^* forms a weighted $w(G)/k$ -separator of G .

Example of the connection

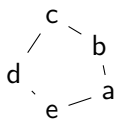


Figure: G .



Figure: D^* .

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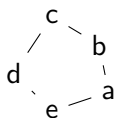


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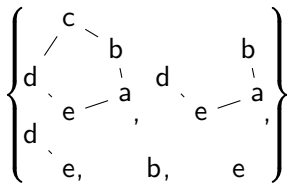


Figure: $\mathcal{R}_{D^*}(G)$.

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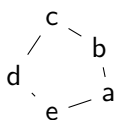


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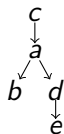


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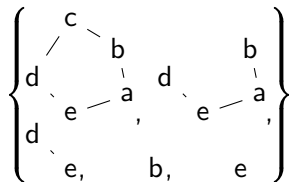


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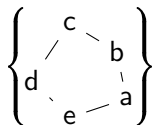


Figure: $\mathcal{L}_5^*, S_5^* = \{\}$.

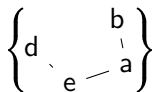


Figure: $\mathcal{L}_4^*, S_4^* = \{c\}$.

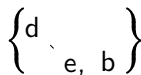


Figure: $\mathcal{L}_3^*, S_3^* = \{a, c\}$.

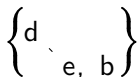


Figure: $\mathcal{L}_2^*, S_2^* = \{a, c\}$.



Figure:



Figure:

$\mathcal{L}_0^*, S_0^* = \{a, b, c, d, e\}$.

Basic lemmas

Lemma

Let $G_{D,v}$ be the candidate subgraph of G in which v is queried when using D . Then, $c_G(D) = \sum_{v \in V(G)} w(G_{D,v}) \cdot c(v)$.

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Proof.

Consider any vertex v . For every $0 \leq k < w(G_{D^*,v})$, $v \notin \bigcup_{H \in \mathcal{L}_k^*} H$, so $v \in S_k^*$ and the contribution of v to the cost is $w(G_{D^*,v}) \cdot c(v)$:

$$\sum_{k=0}^{w(G)-1} c(S_k^*) = \sum_{v \in V(G)} \sum_{k=0}^{w(G_{D^*,v})-1} c(v) = OPT(G).$$

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Lemma

$$2 \cdot OPT(G) = 2 \cdot \sum_{k=0}^{w(G)-1} c(S_k^*) \geq \sum_{k=0}^{w(G)} c(S_{\lfloor k/2 \rfloor}^*).$$

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Lemma

Let $\mathcal{G}$ be any subgraph of G and $0 \leq \beta \leq 1$. Then:

$$\beta \cdot w(\mathcal{G}) \cdot c(S_{\lfloor w(\mathcal{G})/2 \rfloor}^* \cap \mathcal{G}) \leq \sum_{k=(1-\beta)w(\mathcal{G})+1}^{w(\mathcal{G})} c(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{G}).$$

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Theorem

For any $\epsilon > 0$ there exists an $(4 + \epsilon)$ -approximation algorithm for the Tree Search Problem running in $O(n^4/\epsilon^2)$ time.

The algorithm

proc DecisionTree(T, c, w, ϵ):

1. $S_T \leftarrow \text{SeparatorFPTAS}\left(T, c, w, \alpha = 2, \delta = \frac{\epsilon}{4+\epsilon}\right)$.
2. $D_T \leftarrow$ arbitrary partial decision tree for T , built from vertices of S_T .
3. For each $H \in T - S_T$:
 - 3.1 $D_H \leftarrow \text{DecisionTree}(H, c, w, \epsilon)$.
 - 3.2 Hang D_H in D_T below the last query to $v \in N_T(H)$.
4. Return D_T .

Structure of the solution

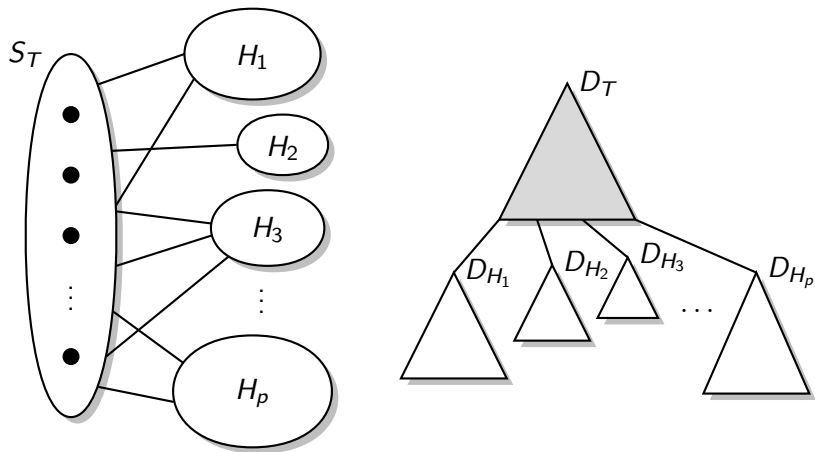


Figure: The separator S_T produced by the algorithm and the structure of the decision tree built using S_T .

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- ▶ Let $\beta = \frac{1-\delta}{2}$. We have: We have that the contribution of the decision tree $D_{\mathcal{T}}$ is bounded by:

$$\begin{aligned} w(\mathcal{T}) \cdot c(S_{\mathcal{T}}) &\leq w(\mathcal{T}) \cdot c(S_{\mathcal{T}}^*) \\ &\leq \frac{2}{1-\delta} \cdot \sum_{k=\frac{1+\delta}{2} \cdot w(\mathcal{T})+1}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{T}\right). \end{aligned}$$

Sketch of the proof

Lemma

$$\sum_{\mathcal{T}} \sum_{k=\frac{1+\delta}{2} \cdot w(\mathcal{T})+1}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{T}\right) \leq \sum_{k=0}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^*\right).$$

► Let D be the resulting decision tree. We have:

$$\begin{aligned} c_{\mathcal{T}}(D) &\leq \frac{2}{1-\delta} \cdot \sum_{\mathcal{T}} \sum_{k=\frac{1+\delta}{2} \cdot w(\mathcal{T})+1}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{T}\right) \\ &\leq \frac{2}{1-\delta} \cdot \sum_{k=0}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^*\right) \leq (4 + \epsilon) \cdot \text{OPT}(T). \end{aligned}$$

► As $1/\delta = \frac{4+\epsilon}{\epsilon} = 1 + 4/\epsilon$, the running time is $O(n^4/\epsilon^2)$.

Min-Ratio Vertex Cut Problem

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Input: Graph $G = (V(G), E(G))$, the cost function $c: V \rightarrow \mathbb{N}$ and the weight function $w: V \rightarrow \mathbb{N}$.

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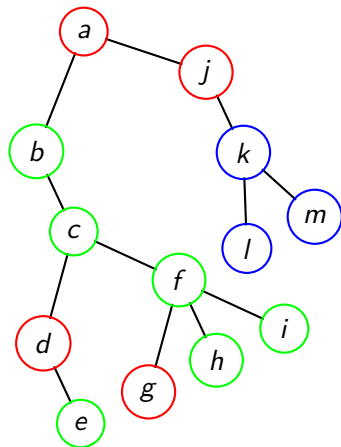
Min-Ratio Vertex Cut Problem

Input: Graph $G = (V(G), E(G))$, the cost function $c: V \rightarrow \mathbb{N}$ and the weight function $w: V \rightarrow \mathbb{N}$.

Output: A partition (A, S, B) of $V(G)$ called **vertex-cut**, such that there are no $u \in A$ and $v \in B$ for which $uv \in E(G)$, minimizing the ratio:

$$\alpha_{c,w}(A, S, B) = \frac{c(S)}{w(A \cup S) \cdot w(B \cup S)}.$$

Example of a vertex cut



	$w(v)$	$c(v)$
a	2	3
b	1	4
c	3	6
d	2	2
e	4	1
f	0	3
g	1	1
h	4	3
i	2	3
j	5	2
k	1	2
l	2	3
m	3	4

Figure: A tree and a vertex cut of ratio

$$\alpha_{c,w}(A, S, B) = \frac{c(S)}{w(A \cup S) \cdot w(B \cup S)} = \frac{8}{(6+10) \cdot (14+10)} = \frac{8}{384} = \frac{1}{48}.$$

How to find the cut

Min-Ratio Vertex Cut Problem is also **NP-hard**. However, we use the following result of [FHL05]:

Theorem

Given a graph $G = (V(G), E(G))$, the cost function $c : V \rightarrow \mathbb{N}$ and the weight function $w : V \rightarrow \mathbb{N}$, there exists a polynomial-time algorithm, which computes a partition (A, S, B) , such that:

$$\alpha_{c,w}(A, S, B) = O\left(\sqrt{\log n}\right) \cdot \alpha_{c,w}(G).$$

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Theorem

Let f_n be the approximation ratio of any polynomial time algorithm for the Min-Ratio Vertex Cut Problem. Then, there exists an $O(f_n)$ -approximation algorithm for the Graph Search Problem, running in polynomial time.

The algorithm

proc DecisionTree(T, c, w, ϵ):

1. $A_G, S_G, B_G \leftarrow \text{AlgorithmMinCut}(G, c, w)$.
2. $D_G \leftarrow$ arbitrary partial decision tree for G , built from vertices of S_G .
3. For each $H \in G - S_G$:
 - 3.1 $D_H \leftarrow \text{DecisionTree}(H, c, w)$.
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Lemma

Let $\mathcal{H} = \mathcal{G} - S_{\mathcal{G}}^$. Then, we can partition $\mathcal{H}$ into two sets, $\mathcal{A}$ and $\mathcal{B}$ such that for $A = \bigcup_{H \in \mathcal{A}} V(H)$ and $B = \bigcup_{H \in \mathcal{B}} V(H)$, we have:*

$$w(A \cup S_{\mathcal{G}}^*) \cdot w(B \cup S_{\mathcal{G}}^*) \geq w(\mathcal{G})^2 / 11.$$

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The partition $(A, S_{\mathcal{G}}^*, B)$ in the above lemma is a vertex cut of $\mathcal{G}$, so we have:

$$\alpha_{c,w}(\mathcal{G}) \leq \frac{c(S_{\mathcal{G}}^*)}{w(A \cup S_{\mathcal{G}}^*) \cdot w(B \cup S_{\mathcal{G}}^*)} \leq \frac{11 \cdot c(S_{\mathcal{G}}^*)}{w(\mathcal{G})^2}.$$

Sketch of the proof

- Let $(A_{\mathcal{G}}, S_{\mathcal{G}}, B_{\mathcal{G}})$, be the partition returned by the algorithm:

$$\alpha_{c,w}(A_{\mathcal{G}}, S_{\mathcal{G}}, B_{\mathcal{G}}) = \frac{c(S_{\mathcal{G}})}{w(A_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}})} \leq f_n \cdot \frac{11 \cdot c(S_{\mathcal{G}}^*)}{w(\mathcal{G})^2}.$$

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- ▶ Assume w. l. o. s. that $w(A_{\mathcal{G}}) \geq w(B_{\mathcal{G}})$.
- ▶ The contribution of the decision tree $D_{\mathcal{G}}$ is bounded by:

$$\begin{aligned} w(\mathcal{G}) \cdot c(S_{\mathcal{G}}) &\leq 11 \cdot f_n \cdot \frac{w(A_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}})}{w(\mathcal{G})} \cdot c(S_{\mathcal{G}}^*) \\ &\leq 11 \cdot f_n \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot c(S_{\mathcal{G}}^*) \\ &\leq 11 \cdot f_n \cdot \sum_{k=w(A_{\mathcal{G}})+1}^{w(\mathcal{G})} c(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{G}) \end{aligned}$$

Sketch of the proof

Let D be the decision tree returned by the procedure. We have:

$$\begin{aligned}c_G(D) &\leq \sum_{\mathcal{G}} w(\mathcal{G}) \cdot c(S_{\mathcal{G}}) \\&\leq 11 \cdot f_n \cdot \sum_{\mathcal{G}} \sum_{k=w(A_{\mathcal{G}})+1}^{w(\mathcal{G})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{G}\right) \\&\leq 11 \cdot f_n \cdot \sum_{k=0}^{w(G)} c\left(S_{\lfloor k/2 \rfloor}^*\right) \\&\leq 22 \cdot f_n \cdot \text{OPT}(G)\end{aligned}$$

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Thank you for your attention!

Questions?

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